Question	Answer	Marks
2(a)(i)	direction of force	**B1**
	force acting on a (test) mass	**B1**
2(a)(ii)	at least four radial lines from the Earth's surface, equally spaced around the surface	**B1**
	arrows indicating direction towards Earth	**B1**
2(b)(i)	top pole labelled S and bottom pole labelled N	**B1**
2(b)(ii)	solenoid field pattern at the poles: at least two field lines either side of both poles, close to the poles, clustered closely together, leaving the surface approximately perpendicularly to the surface and curving away from the axis of the poles as their distance from the surface increases	**B1**
	solenoid field pattern above the equator: at least one field line either side of the Earth connecting two points on the surface that are on the same side of the poles, one north of the equator and one south of it, passing above the surface near the magnetic equator approximately parallel to the surface	**B1**
2(c)(i)	(around the surface) lines are evenly spaced	**B1**
	all lines perpendicular to surface **or** pointing down towards surface (at all points around the surface)	**B1**

Question	Answer	Marks
2(c)(ii)	*Any three bulleted points from:* • strongest at the poles • weakest near the Equator *Up to two points from:* • perpendicular to surface at the poles • parallel to the surface near the Equator • angle to surface increases from Equator to poles	**B3**

Question	Answer	Marks